Bayesian Optimization of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

BYU Ira A. Fulton College of Engineering
Mentored Research Grant Proposal

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Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Duration: 2 Years — **Requested:** $25,000

Abstract

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize multi-material 3D-printed tensegrity-inspired structures for energy absorption. Tensegrity-inspired structures—assemblies of rigid compression members (PLA) and flexible tension members (TPU)—offer tunable mechanical responses and high strength-to-weight ratios. Using a multifidelity Bayesian optimization framework, students will combine finite element simulations (low-fidelity) with physical experiments (high-fidelity) to efficiently navigate the high-dimensional design space. Students will gain hands-on experience in CAD design, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of marketable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of undergraduates: 2–3
Number of graduate students: 1 (co-mentor)

Budget Summary

Category	Year 1	Year 2	Total
Undergraduate wages	$7,500	$7,500	$15,000
Graduate student wages	$2,000	$2,000	$4,000
Supplies	$2,750	$750	$3,500
Travel	$0	$2,000	$2,000
Other	$250	$250	$500
Total	**$12,500**	**$12,500**	**$25,000**

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If the program proves successful, the preliminary data and mentoring framework developed here may inform a future NSF proposal; however, the MRG is self-contained and its outcomes—trained student researchers, conference presentations, and a journal submission—do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target tensegrity-inspired architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU's rate-dependent damping is an *asset* for energy absorption rather than merely a deviation from idealized behavior.

Recent advances in multi-material 3D printing enable the fabrication of tensegrity-inspired structures on a single print platform, using PLA for rigid struts and TPU for flexible tension elements [Ye et al., 2023, Khatri and Egan, 2024]. This eliminates tedious manual assembly and opens a rich, high-dimensional design space. Navigating this space efficiently requires intelligent experimental design; **multifidelity Bayesian optimization** (BO) provides exactly this capability by combining inexpensive simulations (low-fidelity) with physical experiments (high-fidelity) to guide the search toward optimal configurations [Mo et al., 2023, Perdikaris et al., 2017].

The primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Students will gain hands-on skills in CAD, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization. Each student will lead a scoped research project, present at conferences, and co-author a publication.

Figure placeholder: Tensegrity unit cell design → multi-material PLA/TPU printing with wrapping strategy → multifidelity BO loop.

Figure 1: Conceptual overview of the proposed research and mentoring framework.

2 Background

Tensegrity structures consist of isolated compression members (struts) held in equilibrium by a continuous network of tension members [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2015]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact. Multi-material rigid–soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA–TPU combinations printed on a single FDM platform achieve tunable energy absorption. Ye et al. demonstrated a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing expensive blackbox functions using a Gaussian process surrogate [Shahriari et al., 2016, Frazier, 2018]. BO has been applied to materials discovery [Lookman et al., 2019] and metamaterial design [Wang et al., 2022]. Mo et al. [Mo et al., 2023] demonstrated **multifidelity BO** for architected materials, using simulations as a low-fidelity source and experiments as high-fidelity data. We adopt this framework: FEA simulations serve as the low-fidelity source, and 3D-printed tensegrity impact tests provide high-fidelity validation. Because MFBO effectiveness depends on adequate correlation between fidelity levels, we establish a Year 1 Go/No-Go decision gate: students will validate simulation–experiment correlation on a small calibration set before launching full optimization.

3 Student Research Project 1: Simulation and Bayesian Optimization

Student profile: One junior or senior undergraduate with coursework in mechanics of materials or computational methods.

Scope and Deliverables

This student will develop and run the computational pipeline:

1. **Parameterize a tensegrity unit cell** suitable for simulation and multi-material 3D printing, including strut geometry (PLA), tension element geometry (TPU), topology, and boundary conditions.
2. **Build finite element models** using ABAQUS or open-source FEA tools (e.g., FEniCS) to predict force–displacement response and energy absorption under compaction and drop-test loading. These simulations serve as the *low-fidelity* source in the multifidelity framework.
3. **Validate LF–HF correlation (Go/No-Go gate):** Before launching full optimization, the student will run a small calibration set comparing simulated force–displacement curves against experimental results on single unit cells. If the correlation (R^2) between simulated and measured energy absorption is < 0.8, the team will pivot to single-fidelity experimental BO, ensuring feasibility regardless of model accuracy.
4. **Implement a multifidelity Bayesian optimization loop** (following Mo et al. [Mo et al., 2023]) using Python libraries (BoTorch/Ax) to efficiently explore the design space, fusing simulation predictions with experimental results from Student 2.
5. **Identify optimal designs** and generate predictions for experimental validation.

Mentoring outcomes: The student will develop skills in finite element analysis, scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Fabrication, CAD, and Experimental Testing

Student profile: One junior or senior undergraduate with coursework in machine design, CAD, or compliant mechanisms.

Scope and Deliverables

This student will lead fabrication and physical testing:

1. **Design tensegrity-inspired structures in CAD** (SolidWorks or Fusion 360) for multi-material 3D printing, with PLA struts and TPU tension elements printed on a single platform. Following Ye et al. [2023], we adopt a core-wrapping strategy where rigid PLA struts are encapsulated by continuous TPU skins, ensuring robust mechanical interlocking and preventing the interface delamination common in simple layered deposition.
2. **Fabricate structures** using the department's multi-material 3D printer, iterating on print parameters to achieve reliable tensegrity geometries.
3. **Conduct experimental testing:** compaction tests (quasi-static compression) and drop tests (dynamic impact) following the drop-weight protocol of Pajunen et al. [2019]. Primary performance metrics are *specific energy absorption* (SEA) and *compaction efficiency* (ratio of absorbed energy to peak force)—tangible, operationally measurable outcomes suitable for undergraduate research. These experiments serve as the *high-fidelity* source.
4. **Compare experimental results** against simulation predictions from Student 1, quantifying model accuracy and informing surrogate updates.

Mentoring outcomes: The student will develop skills in CAD, additive manufacturing, test design, and data analysis. They will present at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

~~**The mentoring of undergraduate researchers is the central objective of this proposal.** We are committed to providing a structured , supportive environment that develops students into independent , capable researchers~~Each student in this program follows a structured path from guided exploration to independent research ownership, supported by layered mentoring at every stage.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms~~. We target juniors /seniors and will actively recruit from underrepresented groups in engineering~~, targeting juniors and seniors.

Mentoring Structure

- **Weekly individual meetings** (30 min) with Profs. Hill and Baird for technical guidance and professional development.
- **Weekly lab group meetings** (1 hr): progress updates, student literature presentations, skill-building workshops (scientific writing, presentation skills, research ethics).
- **Graduate co-mentor:** Prof. Baird's master's student provides day-to-day guidance in simulation, coding, and experimental technique. This layered model ensures timely support while the graduate student develops mentoring skills.

- **Progressive responsibility:** Students begin with structured tasks and progress to independent design decisions, culminating in ownership of their research project.
- **Peer mentoring:** In Year 2, experienced students mentor newly recruited sophomores, creating a sustainable pipeline of trained researchers.

Student Development Outcomes

Technical risk-mitigation activities are structured as learning modules that develop transferable skills:

- **Technical:** experimental mechanics (test design, data acquisition, interface characterization), computation (FEA modeling, Python, Bayesian optimization), and design (CAD, multi-material 3D printing with wrapping strategies).
- **Communication:** lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript.
- **Professional:** project management, data-driven decision-making (interpreting Go/No-Go correlation results), collaboration with interdisciplinary teams, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

1. **Trained undergraduate researchers:** 2–3 students with hands-on experience in advanced manufacturing, computational modeling, and data-driven optimization, prepared for graduate study or engineering careers.
2. **Conference presentations:** Student-led presentations at **UCUR** (Utah Conference on Undergraduate Research) and **ASME IDETC** (International Design Engineering Technical Conferences).
3. **Journal submission:** One peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*, with undergraduate students as co-authors.
4. **Open-source tools:** Simulation code, optimization scripts, and experimental data published on GitHub for reproducibility.

7 Potential Impact of Work

This project advances architected material design for energy absorption with applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By demonstrating that multifidelity BO can efficiently navigate the tensegrity design space, we establish a generalizable framework for other metamaterial classes. The interdisciplinary training prepares students for graduate study or careers in data-driven design. Open-source dissemination of tools and data extends impact beyond BYU.

Task	Y1 Fall	Y1 Winter	Y2 Fall	Y2 Winter
Student recruitment & onboarding	•			
Literature review	•	•		
Tensegrity parameterization & CAD	•	•		
Interface testing & print validation	•	•		
FEA simulation development		•	•	
LF–HF correlation gate (Go/No-Go)		•		
Multifidelity BO implementation			•	•
Multi-material 3D printing & fabrication		•	•	•
Impact testing & data collection			•	•
Simulation–experiment comparison				•
Conference presentations (UCUR/IDETC)			•	•
Journal manuscript preparation				•
Peer mentoring of new students			•	•

Table 1: Project timeline across four semesters.

8 Timeline

9 Budget

Total requested: **$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Category	Description	Year 1	Year 2
Undergraduate wages	2–3 students, hourly RA	$7,500	$7,500
Graduate student wages	Partial RA for grad co-mentor	$2,000	$2,000
Supplies	Filament, load cells, DAQ, compute	$2,750	$750
Travel	Conference registration & travel	$0	$2,000
Other	Poster printing, consumables	$250	$250
Annual Total		**$12,500**	**$12,500**

Undergraduate wages ($15,000): Supports 2–3 undergraduates for simulation, 3D printing, and testing. **Graduate student wages ($4,000):** Prof. Baird's master's student serves as day-to-day co-mentor. **Supplies ($3,500):** PLA/TPU filament, load cells, DAQ module, fasteners, and cloud compute credits (Year 1 heavy). **Travel ($2,000):** Student presentations at ASME IDETC or UCUR (Year 2). **Other ($500):** Poster printing, consumables, contingency.

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Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	University of Utah
M.S.	Mechanical Engineering	Brigham Young University
B.S.	Mechanical Engineering	Brigham Young University

Appointments

- Associate Professor, Department of Mechanical Engineering, Brigham Young University (current)
- Research Engineer, Sandia National Laboratories (prior)

Products Most Closely Related to This Proposal

Publications and products related to smart materials, shape memory alloys, compliant mechanisms, and structural mechanics.

Synergistic Activities

- Mentoring of undergraduate and graduate students in experimental mechanics and smart materials research at BYU.
- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with national laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah
B.S.	Materials Science and Engineering	Brigham Young University

Appointments

- Faculty, Department of Mechanical Engineering, Brigham Young University (current)

Products Most Closely Related to This Proposal

Publications and products related to materials informatics, Bayesian optimization, data-driven materials discovery, and machine learning for materials science.

Synergistic Activities

- Development of open-source tools for Bayesian optimization and materials informatics.
- Mentoring of graduate and undergraduate students in data-driven methods for materials science and engineering.
- Active contributions to the materials informatics community through publications, software, and educational resources.